

Topological Analysis of Protein Lasso Structures Using Persistent Homology

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Abstract

Lasso motifs are entangled protein structures in which a covalently closed loop is pierced by one or both termini of the polypeptide chain. Detecting such motifs is typically based on geometric methods involving minimal surfaces or the Gaussian linking number. In this work, we propose a new approach based on persistent homology, which captures the interaction between the loop and the protein tail through changes in persistence diagrams. By measuring bottleneck distances as tail atoms are incorporated into the loop structure, we identify piercing events and automatically detect lasso intersections. We evaluate the method on 4,846 proteins with 58,564 lassos from the LassoProt database and show strong agreement with the minimal surface approach. We further analyze highly supercoiled proteins from the AlphaLasso dataset derived from AlphaFold predictions and demonstrate that our method reliably detects supercoiling patterns. Persistent homology thus provides a stable and computationally efficient method for analyzing lasso topologies in proteins.

1 Introduction

Proteins are chain macromolecules composed of α -amino acid residues folded into complex three-dimensional structures. These α -amino acid residues are connected by peptide bonds forming the protein backbone, but each residue also has a side chain that distinguishes different α -amino acid types. Interactions between amino acid residues and the environment determine the structure and dynamics of a protein, which are crucial for its function and stability. The process by which a protein obtains its native (functional) structure is called protein folding. The standard way of describing protein structure divides it into four levels of organization:

1. primary structure – sequence of amino acids along the backbone (Figure 1a);
2. secondary structure – local conformation of an amino acid sub-chain: helix, β -strand, and turn;
3. tertiary structure – conformation of a single amino acid chain, including covalent bonds between sequentially non-adjacent residues and the spatial organization of secondary structures;
4. quaternary structure – full structure of a multi-chain protein, including covalent and non-covalent interactions between amino acid chains.

This standard four-level view of protein structure overlooks protein topological complexity. A protein without covalent bonds between side chains is topologically a strand (or a set of strands), see Figure 1b.

Each additional covalent bond, typically formed by a disulfide bridge or macrolactam bond, changes the topology of the protein, see Figure 2. Intra-chain bonds create a topological loop that can be pierced by the remaining parts of a chain (tails), see Figure 2b. Such pierced loops are called lassos and are studied in this article.

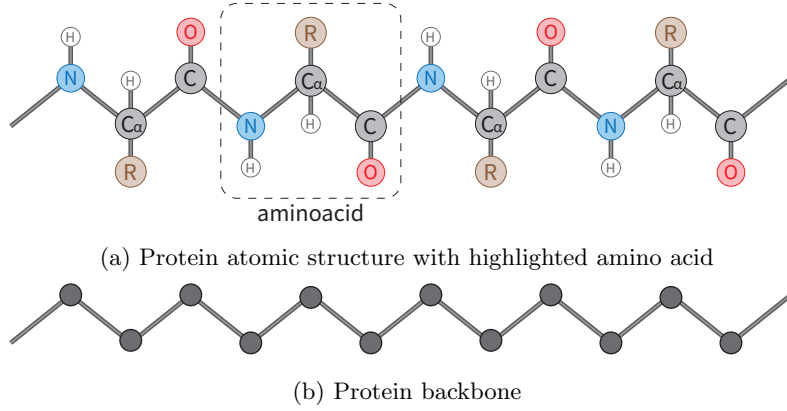


Figure 1: Atomic and backbone views of a protein. (a) Atomic structure with a residue “R” and a highlighted amino acid. (b) Backbone representation using the N , C_α , and C atoms of the residue.

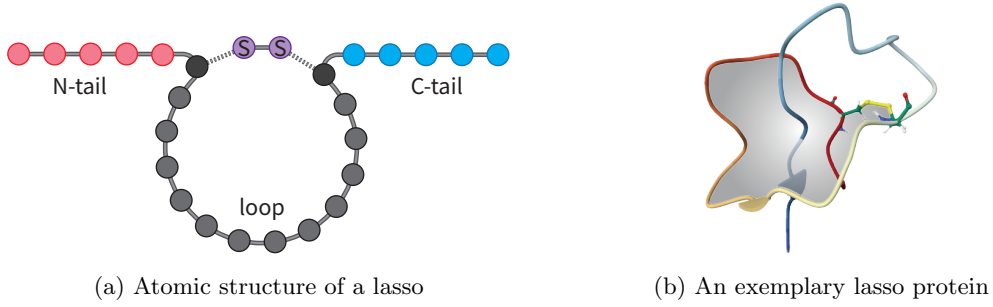


Figure 2: Representation of a lasso protein. (a) Atoms belonging to the N-tail, loop, and C-tail; the loop is closed by a cystine bond (SS bond). (b) Left: an exemplary lasso protein (PDB code 2MXQ) of type L_1 and the corresponding minimal surface.

In [10], six distinct lasso types were classified, revealing the structural diversity of these entangled motifs. The loop refers to the closed backbone segment, while the N-tail and C-tail are the respective termini on the amino- and carboxy-terminal sides of the loop.

Following Figure 3, the simplest class of lassos is denoted L_n , where a single tail pierces the minimal surface exactly n times. Thus, L_0 corresponds to a trivial lasso, while L_1 , L_2 , and L_3 represent loops pierced once, twice, or three times by a tail, respectively.

A more complex class is the supercoiling lasso, denoted LS_n , where the same tail pierces the minimal surface n times while winding around the loop in a supercoiled manner.

The minimal surface has two oriented sides, which we denote by the positive (+) and negative (−) side. Each intersection is assigned a sign depending on the direction in which the tail crosses the surface.

We therefore use the notation $LS_{n\sigma T}$ to describe the full topology of a lasso motif. Here n denotes the number of piercings, σ is the sequence of intersection signs, and $T \in \{C, N\}$ indicates whether the piercing tail originates from the C- or N-terminus of the protein. For example, LS_{2++C} denotes a supercoiled lasso where the C-tail pierces the loop twice from the positive side, while LS_{3++-C} indicates three C-tail piercings with signs +, +, − (Figure 3).

The study of lasso proteins presents both mathematical and computational challenges, as protein chains are open curves, making their topological classification complex. In particular, lassos (and slip-knots) require distinguishing between topological and geometric constraints, where the presence of entanglement significantly affects protein stability and function. A lasso motif is topologically trivial because the tail can, in principle, be isotoped out of the loop. However, it remains geometrically nontrivial, as the

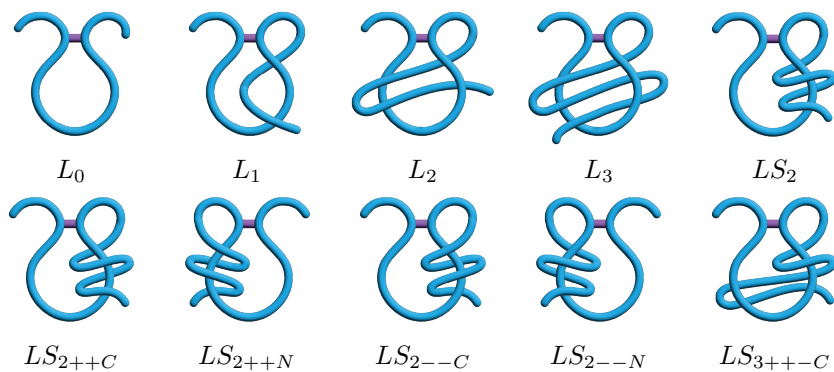


Figure 3: A schematic depiction of representative lasso motifs.

tail physically pierces the loop, constraining its conformational flexibility. This distinction necessitates specialized methods for detection and classification.

Two computational techniques have been developed to detect and classify lasso topology. The minimal surface approach, introduced in [10], constructs a spanning surface over the macrolactam ring and determines whether the peptide tail passes through it. The number of piercings across this surface classifies the lasso type. Later, the computational complexity of this method was greatly reduced by computing the Gauss linking number (GLN), which provides an efficient algebraic measure of loop-tail entanglement [9].

Given the limitations of purely topological methods, persistent homology (PH) offers a complementary approach. PH detects topological properties of spaces but relies heavily on their geometric presentation, making it particularly suitable for capturing the structural constraints of lasso proteins. Unlike classical discrete methods, PH provides a stable multi-scale perspective of the protein’s entanglement, allowing for a more nuanced characterization of its folding dynamics. In a nutshell, PH represents geometric sizes of the components and holes in a space as they appear across multiple scales. A particular advantage of PH, as opposed to other discrete invariants, is its stability: geometrically similar inputs yield similar persistent homology. This structural property makes PH a stable descriptor of geometric shapes.

By using persistent homology, we study lasso proteins in comparison with the minimal surface method and investigate how their geometric and topological constraints affect the folding process.

2 Methods: Persistent Homology

2.0.1 Reasoning and geometric intuition

We propose a novel approach based on persistent homology (PH). While the existing method using minimal surfaces has been highly effective, our approach offers a different perspective with improved computational efficiency and stability.

We first discuss the geometric intuition and structural properties of both methods. A simplified geometric setting is the following: given a loop in three-dimensional space and an attached tail (thread) emanating from it at some point, determine whether, and to what degree, the thread pierces the initial loop, see Figure 3.

The minimal surface method approaches the problem as follows. It spans a minimal surface along the initial loop, with this surface representing the smallest-area geometric interior of the loop. The method then computes the number and orientation of the intersections of the tails with the spanned surface. While this method has been highly successful, two aspects motivate an alternative viewpoint. First, the computation of a minimal surface is computationally demanding, as it requires solving the Plateau problem. Second, a minimal surface need not be unique. Figure 7 depicts two different minimal surfaces with the same boundary. While such non-uniqueness may occur only for certain non-generic loops, it indicates that the resulting piercing description may depend sensitively on the chosen spanning surface. In particular, small perturbations of the loop or tail can induce abrupt changes in the detected intersections.

To address these issues, we use PH as follows. The main loop in our geometric setting can be detected by topological methods, in particular by homology, as a one-dimensional hole. Now consider increasingly thickened offsets of the loop. At some scale, call it the terminal scale $\mathcal{T}$, the offset becomes large enough

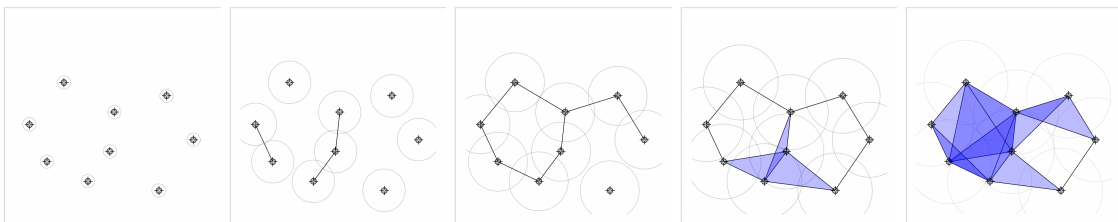


Figure 4: Nine points in the plane with growing offsets indicated by circles, together with the corresponding simplicial complexes superimposed. PH tracks how components merge and how holes appear and are eventually filled.

to fill in the hole, a structural change detectable by homology. Now assume that, besides thickening the loop, we also add a single point x in space. Our guiding geometric principle is that the parameter $\mathcal{T}$ decreases the most when x pierces the original loop. The advantages of this approach are twofold. First, the relevant parameters can be computed efficiently and to high precision using standard PH software. Second, the degree to which x pierces the loop is encoded by a continuous quantity. Adding a point far from the loop will not change $\mathcal{T}$ appreciably, while adding a point near the center of a circular loop will decrease $\mathcal{T}$ significantly; see Figure 9.

2.0.2 PH formally

In this section, we briefly review the persistent homology models used in our analysis and their applications in biomolecular data analysis.

Homology is a discrete topological invariant of spaces that measures connectivity. It is one of the foundational tools of topology. Zero-dimensional homology gives the number of connected components of a space, while one-dimensional homology gives the number of holes; higher-dimensional variants are beyond the scope of this paper. Motivated by the protein docking problem, persistent homology [7, 6] was developed at the turn of the millennium as a stable [3], parameterized, and efficiently computable version of homology suitable for geometric data analysis. It has since become a powerful tool in data analysis, with applications across the sciences [5] and in biology in particular [12, 13]. Motivated by the use of α -shapes to model molecular structures geometrically [1, 8], the original geometric idea of PH was to track the number of holes in growing offsets. An example of the union of growing disks is shown in Figure 4. There we also see a combinatorial structure called a simplicial complex encoding the union of the balls. It consists of vertices, edges, and triangles, and contains sufficient information to compute efficiently the number of components and holes in the offset. The growth of a simplicial complex is performed by adding new edges and triangles. Persistent homology of a growing simplicial complex is represented in two equivalent ways, as shown in Figure 5: either as a barcode, with bars indicating the lifespan of components or holes, or as a persistence diagram, with points in the plane representing the endpoints of the bars. While Figure 5 represents PH in dimension 0, Figure 6 represents PH in dimension 1.

In our approach, we use specific simplicial complexes called Vietoris–Rips complexes, where the vertices X consist of the selected carbon and nitrogen atoms. Given a scale $r \geq 0$, the Vietoris–Rips complex of X consists of all edges and triangles for which the pairwise distance between the defining points is at most r . When computing PH, practitioners typically use Vietoris–Rips complexes because of their computational efficiency [2], while theoretical guarantees ensure that the output closely resembles what would be obtained by modeling the offsets of the vertices directly. The latter can be modeled exactly by α -complexes or Čech complexes, but at a much greater computational cost [2, 6, 14].

As indicated above, one of the main advantages of PH is its stability with respect to perturbations of the input [3]. In our setting, this takes the following form. Fix a vertex set X in space and choose a perturbation parameter $\varepsilon > 0$. Now perturb X so that each vertex moves by at most ε , obtaining a new vertex set X' . Then the barcodes obtained from the Vietoris–Rips filtrations of X and X' differ by at most ε in the sense that the endpoints of each bar are perturbed by at most ε . Thus, a continuous change in X leads to a continuous change in the barcode intervals.

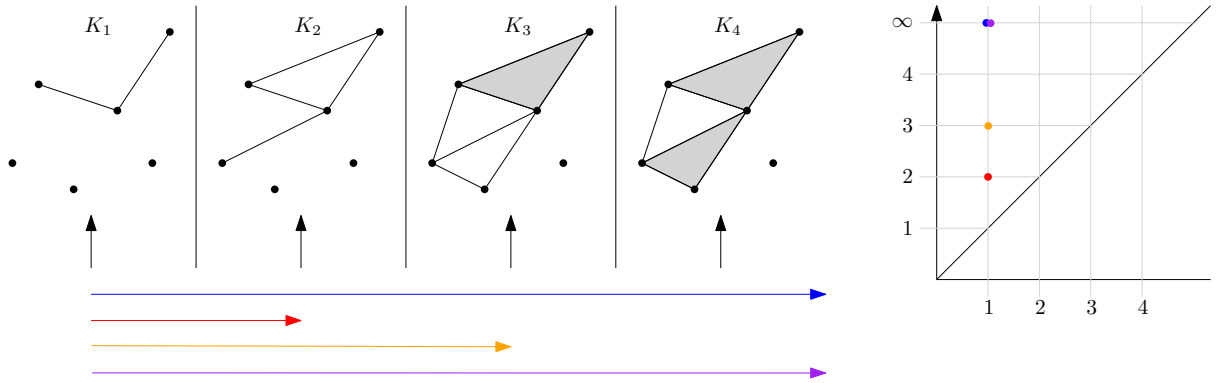


Figure 5: PH in dimension 0, figure adapted from [14]. The bars indicate the lifespans of connected components. Having two bars $[1, \infty)$ means that the corresponding point in the persistence diagram has multiplicity two.

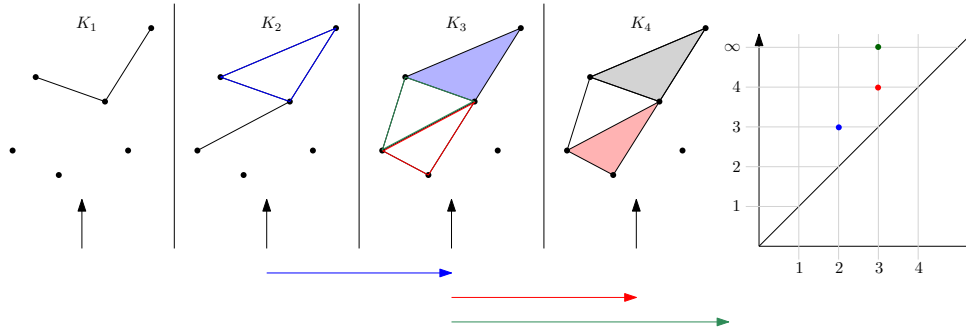


Figure 6: PH in dimension 1, figure adapted from [14]. The bars indicate the lifespans of holes. The blue bar arises when the hole indicated by the red edges appears, and it terminates when the hole is filled by the red triangle. In general, a hole may require several triangles to be filled.



Figure 7: Non-stability of the minimal surface description.

2.1 Detecting lassos through persistent homology

Each lasso consists of a loop closed by a cystine bridge and at most two tails, the N-tail and the C-tail. For our base model of the loop, we use the 3D coordinates of the backbone atoms within the loop, specifically the repeating $-N - C_\alpha - C' -$ units spanning the segment between the α -carbons of the cysteine residues that form the disulfide bond, together with the cystine bond itself. We treat this sequence of loop atoms as

$$L = (S_i, C_i^\beta, C_i^\alpha, C'_i, N_{i+1}, C_{i+1}^\alpha, C'_{i+1}, \dots, N_{j-1}, C_{j-1}^\alpha, C'_{j-1}, N_j, C_j^\alpha, C_j^\beta, S_j).$$

Next, we take the 3D coordinates of the repeating $-N - C_\alpha - C' -$ units of the N-tail from the N-terminus up to the nitrogen atom of the cysteine residue:

$$T_N = (N_1, C_1^\alpha, C'_1, N_2, C_2^\alpha, C'_2, \dots, N_{i-1}, C_{i-1}^\alpha, C'_{i-1}, N_i).$$

Similarly, we define the C-tail from the carboxylic carbon of the cysteine residue to the C-terminus:

$$T_C = (C'_j, N_{j+1}, C_{j+1}^\alpha, C'_{j+1}, \dots, N_n, C_n^\alpha, C'_n).$$

To detect whether the N-tail pierces the surface spanned by L , we compute the persistence diagram $\mathcal{D}(L)$ and the diagrams $\mathcal{D}(L \cup \{x\})$ for each $x \in T_N$. We then compute the bottleneck distances

$$d_B(\mathcal{D}(L), \mathcal{D}(L \cup \{x\})).$$

In an ideal scenario, the bottleneck distance increases when x is a tail atom that pierces the region enclosed by L , as illustrated in Figure 8.

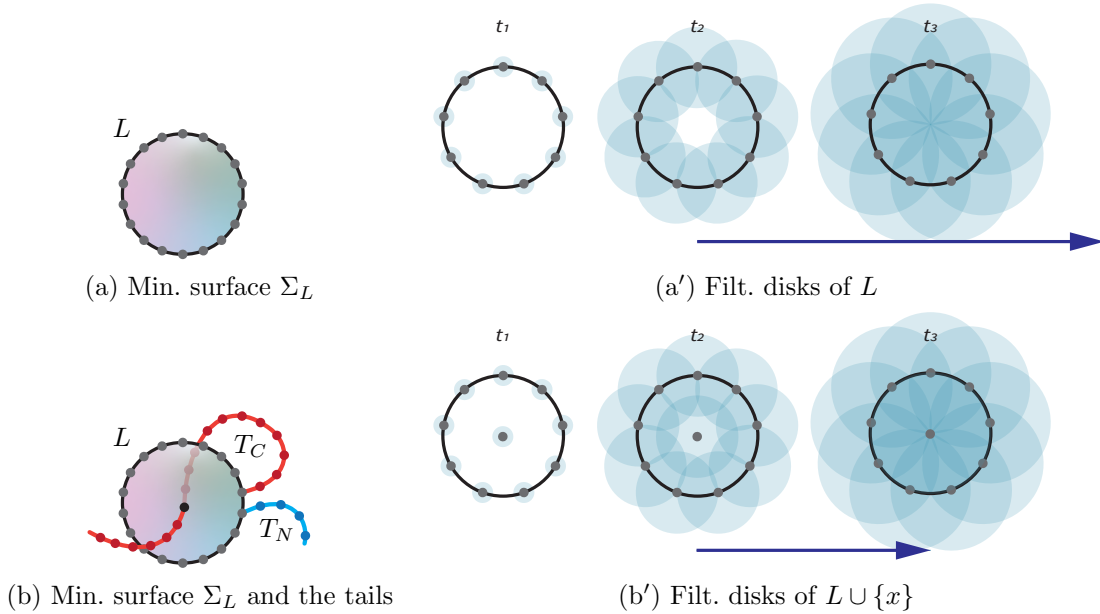


Figure 8: The top row presents: (a) the minimal surface corresponding to a loop, and (a') the associated point cloud with filtration disks at times $r = t_1$, $r = t_2$ (birth), $r = t_3$, and $r = t_4$ (death). The bottom row illustrates the same setup with an additional atom x (black): (b) the minimal surface of the loop together with x , and (b') the corresponding point cloud with filtration disks at times $r = t_1$, $r = t_2$ (birth), $r = t_3$ (death), and $r = t_4$. In this case, the death occurs sooner than in case (a). If x is a puncture point, the persistence pair in $\mathcal{D}(L)$ should exhibit a shorter lifespan in $\mathcal{D}(L \cup \{x\})$.

The bottleneck loop diagram is a plot in which the horizontal axis represents the position along the tail and the vertical axis represents the bottleneck distance, measuring the topological change between the loop and the loop together with a tail atom x (see the right-hand panels in Figure 10). Local maxima in this diagram indicate positions where the bottleneck distance changes significantly and hence correspond to transitions detected by persistent homology.

From Figure 10, we observe a clear pattern for simple lasso types: an L_n lasso is characterized by n local maxima.

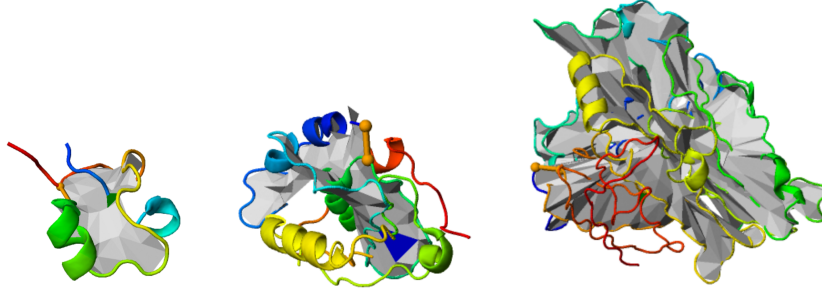


Figure 9: Examples of proteins with L_1 -type lassos of varying geometric complexity. From left to right: a simple lasso (PDB 6VID), a moderately complex lasso (PDB 6Q10), and a geometrically complex lasso (PDB 6KLG).

Because lasso motifs are typically non-circular in geometry (see Figure 9), the bottleneck distances fluctuate irregularly as tail atoms approach the loop. To mitigate this irregularity, we apply a sequence of filtering steps to smooth the bottleneck distance function: (i) thresholding (values below 0.25 are removed), (ii) moving average smoothing with a window of size 17, (iii) second thresholding (values below 20% of the function range are removed), and (iv) peak cutoff (neighboring values are merged if their difference is below 20% of the function range). We define a PH intersection as the midpoint of a plateau in the filtered bottleneck distance function, marking a significant transition in persistent homology. These parameter values were determined experimentally so as to maximize the F_1 score for the proteins contained in LassoProt, the database of proteins with lassos [4].

Determining the lasso type. To distinguish between non-supercoiled and supercoiled lassos, for example L_2 and LS_2 , we use a local linear-algebraic criterion.

If a tail T pierces the loop L at a point $x_i = T(i) \in T$, we define $y_j = L(j)$ as the closest point in L to x_i . To analyze the local geometry, we consider the neighboring points of y_j and x_i :

$$\begin{aligned} N(y_j) &= (L(j-2), L(j-1), L(j), L(j+1), L(j+2)) \subset L, \\ N(x_i) &= (T(i-2), T(i-1), T(i), T(i+1), T(i+2)) \subset T. \end{aligned}$$

We compute principal vectors $\vec{s}_L$ and $\vec{s}_T$ using principal component analysis (PCA) applied to the point sets $N(y_j)$ and $N(x_i)$, respectively.

The sign of the mixed product $(\vec{s}_L, x_i - y_j, \vec{s}_T)$ determines the local side from which the tail pierces the loop. For supercoiled lassos, these signs tend to remain constant across successive piercings, whereas for non-supercoiled lassos they alternate. This criterion allows us to distinguish between the two cases and to assign intersection signs.

2.2 Results

Comparison of intersection point detection using persistent homology and the minimal surface method. We compare the detection of intersection points in lasso proteins using persistent homology (PH) and the minimal surface method (MSM) from LassoProt [4]. The results are summarized in Table 1.

One main difference between our approach and the one used in LassoProt is that we include all backbone atoms relevant to the construction of the loop, rather than only α -carbons as in [4]. This denser representation makes the loop structure more pronounced and allows more precise identification of lasso topology.

We compare the intersection points detected by PH with those reported in LassoProt. An intersection is considered correctly detected if its position differs by at most D amino acids from the reference intersection, where $D \in \{1, 3, 6, 9, 12\}$. We computed intersections for 58,564 lassos, including trivial ones, from 4,846 proteins.

Performance is evaluated using precision, recall, and the F_1 score:

$$F_1 = \frac{2PR}{P + R},$$

where P denotes precision and R denotes recall.

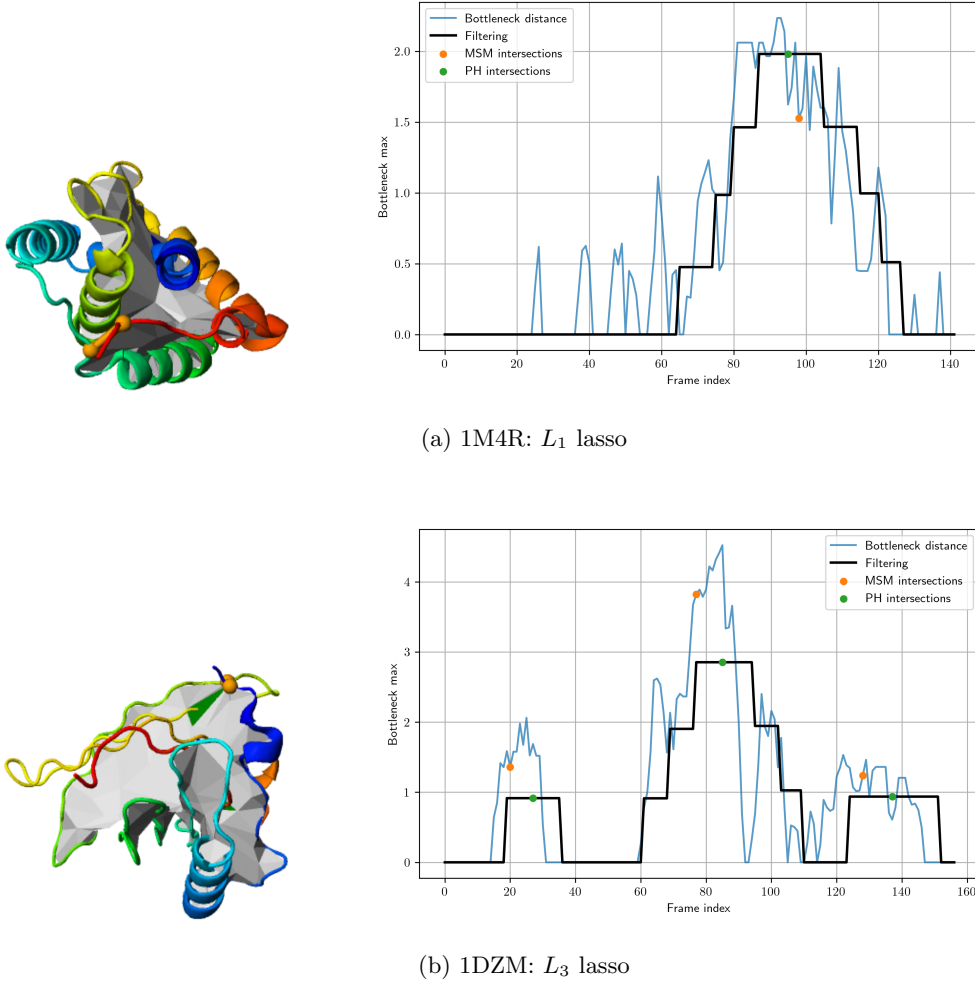


Figure 10: Examples of proteins and their bottleneck distance curves illustrating different lasso types.

2.3 Applications

Detection of lasso motifs in protein folding pathways. We apply PH to analyze folding pathways of lasso motifs through molecular dynamics (MD) simulations.

As an example, we consider the protein 1ZD0 from *Pyrococcus furiosus*, which exhibits lasso-like configurations in its native state. To investigate its folding behavior, we start from an unfolded state with native contacts and track the conformational changes during the simulation. For each frame, we compute the maximal bottleneck distance relative to the initial unfolded frame, thereby quantifying structural deviations and tracking the emergence of lasso motifs. Figure 11 shows that PH detects the formation of the lasso during the simulation.

Supercoiling in protein folding pathways. The dynamics of the Pfu-542154 protein was analyzed using a C_α coarse-grained structure-based model in [11], where the minimal surface approach was used to characterize the folding pathway.

In Figure 12, we present the folding pathway analysis of Pfu-542154 during an MD simulation, where the protein transitions from an unfolded state to a supercoiled state. The horizontal axis represents the frame index of the simulation. The blue scatter points correspond to the number of maxima in the bottleneck distance diagram of the loop structure. At early stages, there are no maxima, reflecting an unfolded conformation. As the simulation progresses, the first piercing event occurs, and in the supercoiled state the plot stabilizes at two maxima, corresponding to the two piercings that define the final topology.

A key distinction between these methods lies in the detection of transitional states. In the minimal

Table 1: Performance of the persistent homology method in detecting intersection points compared to the minimal surface method implemented in LassoProt. Detection is considered correct if the PH intersection lies within a given number of atoms from the reference intersection. The analysis was performed on 4,846 proteins from the LassoProt database.

Distance D (atoms)	Precision	Recall	F_1 score
1	0.508	0.669	0.577
3	0.579	0.750	0.653
6	0.653	0.828	0.730
9	0.699	0.869	0.775
12	0.714	0.877	0.787

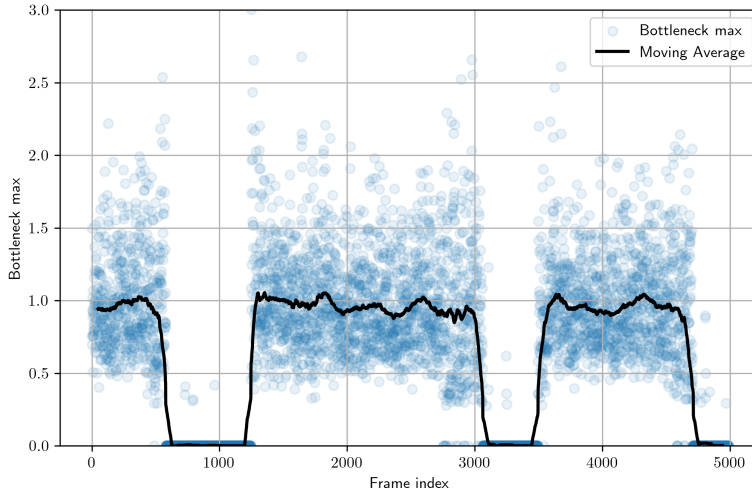


Figure 11: An MD simulation of protein 1ZD0 from an unfolded state to a folded state. The horizontal axis represents the simulation frame, and the vertical axis shows the maximal bottleneck distance in that frame. The onset of lasso formation is visible when the moving average rises above the threshold.

surface approach, the formation and threading of the loop are described explicitly, whereas in PH the transient single-piercing intermediate is only weakly visible. This suggests that during the MD simulation the loop adopts a U-shaped conformation and the tail threads through the loop on a very short timescale.

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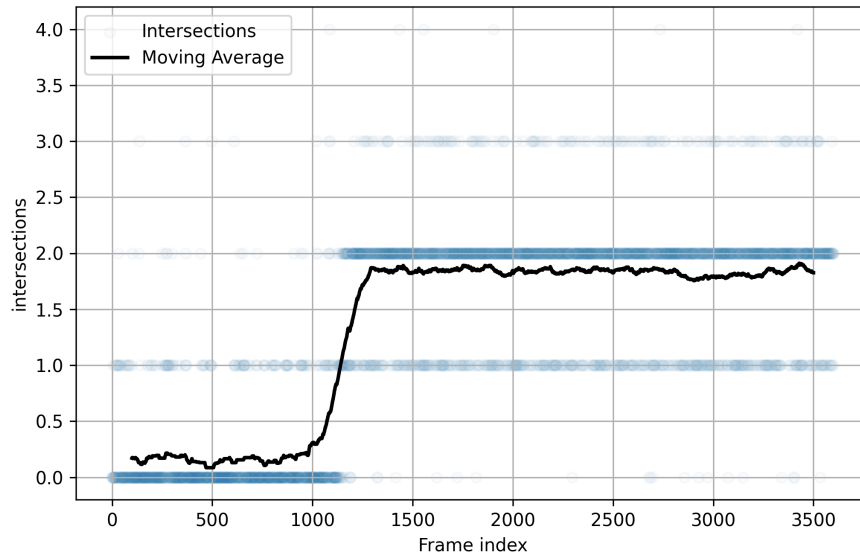


Figure 12: Visualization of the folding dynamics of a supercoiled lasso protein during an MD simulation. The horizontal axis represents the frame index, while the blue scatter points indicate the number of maxima in the bottleneck distance diagram of the loop. The line represents the moving average and highlights the transition from the unfolded to the supercoiled state.

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